

PROJECT SUBMISSION REPORT

Online Travel Booking & Real-Time Flight Tracking Platform

Course / Track:	Full-Stack Web Development Capstone Project
Developer:	Shagun Kaushal Prajapati
Technologies:	Next.js 16 (Turbopack), React 19, TypeScript, Tailwind CSS, Recharts
Submission Date:	August 28, 2026
Live Web App:	https://makemytrip-one.vercel.app
GitHub Repository:	https://github.com/shagun201104/makemytrip

1. INTRODUCTION & PROJECT OVERVIEW

This project is a full-stack web platform for travel bookings, modeled after MakeMyTrip. The goal was to build a clean, working web application where users can search for flights and hotels, track live flight status updates in real time, view dynamic pricing trends, select specific seats or hotel rooms, and manage cancellations with refund calculations.

The application is built using Next.js 16 (App Router) with TypeScript and Tailwind CSS. State management is handled using React hooks and browser local storage for persistence across page reloads.

1.1 Key Objectives

- 1. Implement a search system for flights and hotels with input validation.
- 2. Build a live flight tracking engine simulating status changes (On Time, Boarding, Delayed, Landed) based on actual time.
- 3. Develop a dynamic pricing module adjusting fares according to peak travel periods and weekend demand.
- 4. Design interactive seat and room selection grids for booking customization.
- 5. Create a self-service cancellation dialog calculating refund amounts based on timing.
- 6. Ensure full mobile responsiveness across all devices, including iOS and Android browsers.

2. SYSTEM DESIGN & ARCHITECTURE

The application follows Next.js App Router file-based routing structure. Client-side components handle user interaction, while state is stored locally or processed via helper functions.

2.1 Directory Layout

```
makemytour/  
|-- app/  
|   |-- layout.tsx           # Root layout & navbar  
|   |-- page.tsx             # Homepage search card & showcase  
|   |-- flight-status/page.tsx # Live flight tracking dashboard  
|   |-- admin/page.tsx       # Admin control panel  
|   |-- booking/              # Flight & hotel checkout maps  
|   |-- lib/flightStatusApi.ts # Flight status engine  
|-- components/  
|   |-- PriceHistoryChart.tsx # Recharts price visualization  
|   |-- LivePrice.tsx         # Dynamic price calculator  
|   |-- booking/              # Seat maps, room grids & cancellation dialogs
```

3. IMPLEMENTATION DETAILS BY MODULE

3.1 Flight & Hotel Search Engine (app/page.tsx)

The homepage contains a dual-tab search card for flights and hotels. Users can select departure/destination cities, departure date, and number of passengers using steppers (+ and - controls).

- Input Validation: Ensures departure and destination cities are not identical and mandatory fields are filled.
- Mobile Layout Fix: Wrapped `<input type='date'>` elements inside a wrapper `<div>` styled with `overflow-hidden border-2 border-[#cbd5e1] rounded-2xl min-w-0 w-full` to resolve WebKit border clipping.

3.2 Live Flight Status Tracking (app/flight-status/page.tsx)

The flight tracking page simulates live status changes for active flights.

- Time Calibration: Engine runs at a 1:1 real-time scale (1 second = 1 second of flight time).
- Status Updates: Flights move through states (ON TIME -> BOARDING -> IN FLIGHT -> LANDED or DELAYED).
- Management: Supports custom flight number tracking and 1-click deletion with localStorage persistence.

3.3 Dynamic Pricing Engine (components/LivePrice.tsx & PriceHistoryChart.tsx)

Base prices automatically recalculate based on demand metrics:

1. Peak Season Multiplier: Fares increase by 20% during holiday dates.
2. Weekend Surge: Fares increase by 10% on Friday through Sunday departures.
3. Price History Graph: Interactive line chart built with Recharts shows 7-day price trends.

3.4 Interactive Seat & Room Selection

- Flight Seat Map (InteractiveSeatMap.tsx): Displays Business and Economy cabin grid with occupied/available seat pricing.
- Hotel Room Grid (InteractiveRoomGrid.tsx): Displays room types (Deluxe, Suite, Villa) with occupancy limits and amenity tags.

3.5 Cancellation & Refund Handling (CancellationDialog.tsx)

Allows users to cancel existing bookings and view a breakdown of their refund:

- Cancellation > 24 hours before trip: 100% refund (minus standard transaction fee).
- Cancellation <= 24 hours before trip: 80% refund.

3.6 Admin Panel (app/admin/page.tsx)

Dashboard styled with a dark navy background (#0f172a) to manage platform stats, monitor active flights, and check revenue figures.

4. CHALLENGES & TECHNICAL RESOLUTIONS

Problem Encountered	Technical Root Cause	Resolution Applied
Date input overflow on mobile	iOS Safari WebKit date controls render with fixed intrinsic width.	Wrapped input in <code>min-w-0 overflow-hidden rounded-2xl border-2</code> container with <code>appearance-none</code> .
Fast simulation speed	Flight status was updating every few seconds, feeling unrealistic.	Recalibrated timer ticks in <code>flightStatusApi.ts</code> to 1:1 real-time scale.
Deleted flights re-appearing	State updated locally but was not syncing to localStorage.	Updated delete handler to sync updated array to <code>localStorage.setItem()</code> immediately.
Button color mismatch	Search buttons used inconsistent colors across pages.	Unified all search action buttons to #215190 (Steel Sapphire Navy) with 44px height.

5. TESTING AND BUILD RESULTS

The project was compiled and tested using `npx next build`. All pages compile cleanly as static prerendered routes.

Next.js 16.2.12 (Turbopack) Compiled successfully in 1373ms Running TypeScript ... Passed in 2.0s Generating static pages (9/9) in 131ms		
Route (app)	Size	First Load JS

+ /	182 kB	284 kB
+ /_not-found	982 B	102 kB
+ /admin	38.4 kB	140 kB
+ /booking/flight	24.1 kB	126 kB
+ /booking/hotel	22.8 kB	125 kB
+ /flight-status	31.6 kB	133.6 kB
+ /profile	12.4 kB	114.4 kB

6. CONCLUSION

The MakeMyTrip clone application fulfills all core functionality requirements. The platform provides a clean user interface, accurate flight status tracking, dynamic pricing feedback, and responsive layout performance across desktop and mobile screens.

The application is deployed live on Vercel at <https://makemytrip-one.vercel.app>.